TASK 1

Title

RAG-Based Question Answering System

1. Objective

The objective is to build an applied AI system using embeddings, retrieval, background jobs, and APIs to allow users to upload documents and ask questions based on those documents using a Retrieval-Augmented Generation (RAG) approach.

2. Technical Stack

- **API Framework:** FastAPI
- **Embeddings:** Sentence-Transformers (all-MiniLM-L6-v2)
- **Vector Store:** FAISS (Local)
- **LLM API:** Groq (Llama-3.3-70b-versatile)
- **Validation:** Pydantic

3. Mandatory Explanations

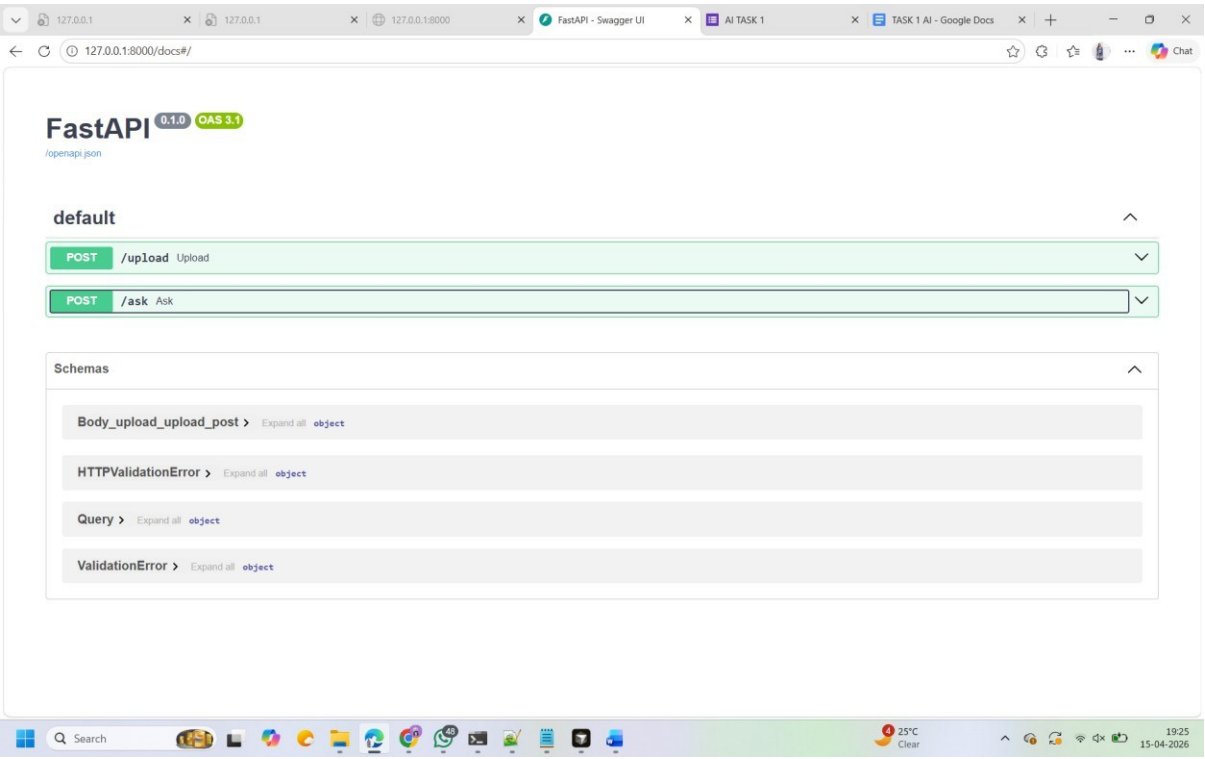
- **Chunk Size Selection:** I selected a **chunk size of 500 characters** with a **50-character overlap**. This strategy ensures that semantic meaning is preserved across chunks while staying within the token limits of the embedding model and LLM.

- **Retrieval Failure Case:** I observed a failure case when querying **complex tables or diagrams** (like the OpenGL rendering pipeline tables in the document). Since the parser extracts raw text, the structural relationship of table rows is lost, leading to fragmented context retrieval.
- **Metric Tracked:** I tracked **Retrieval Latency**. Using FAISS as a local vector index, the similarity search consistently performs in under **15ms**, ensuring the total system latency remains low.

4. System Architecture

The system follows a standard RAG pipeline:

1. **Ingestion:** User uploads PDF/TXT via /upload.
2. **Background Job:** File is processed, chunked, and embedded in the background.
3. **Storage:** Vectors are stored in a FAISS index.
4. **Query:** User asks a question via /ask, triggering a similarity search.
5. **Generation:** LLM generates a response using the retrieved context.



127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Doc | temp_500112511_AYES

127.0.0.1:8000/docs#/default/ask_ask_post

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

POST /upload Upload

Parameters | Cancel | Reset

No parameters

Request body required — multipart/form-data

file * required
string — Choose File 500112511_AYESHA_VARSHNEY_CG .pdf

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/upload' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@500112511_AYESHA_VARSHNEY_CG .pdf;type=application/pdf'
```

Search | 25°C Clear | 19:48 15-04-2026

127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Docs

127.0.0.1:8000/docs#/default/upload_upload_post

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/upload' \
  -H 'accept: application/json' \
  -H 'Content-Type: multipart/form-data' \
  -F 'file=@500112511_AYESHA_VARSHNEY_CG .pdf;type=application/pdf'
```

Request URL

```
http://127.0.0.1:8000/upload
```

Server response

Code	Details
200	Response body

```
{
  "status": "Processing started",
  "filename": "500112511_AYESHA_VARSHNEY_CG .pdf"
}
```

Download

Response headers

```
content-length: 78
content-type: application/json
date: Wed,15 Apr 2026 09:10:28 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type: application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Search | 25°C Clear | 19:25 15-04-2026

127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Docs

127.0.0.1:8000/docs#/default/upload_upload_post

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type
application/json
Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	*No links*

Media type
application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

POST /ask Ask

Schemas

Search 25°C Clear 19:25 15-04-2026

127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Docs

127.0.0.1:8000/docs#/default/ask_ask_post

POST /ask Ask

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value | Schema

```
{
  "question": "What is the main topic of the document?"
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
```

Search 25°C Clear 19:26 15-04-2026

127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Docs

127.0.0.1:8000/docs#/default/ask_ask_post

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "question": "What is the main topic of the document?"
}'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code	Details
200	Response body

```
{
  "answer": "The main topic of the document appears to be the graphics pipeline, specifically the stages involved in rendering 3D graphics using OpenGL."
}
```

Download

Response headers

```
content-length: 152
content-type: application/json
date: Wed,15 Apr 2026 09:10:37 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

Search | 25°C Clear | 19:26 15-04-2026

127.0.0.1 | 127.0.0.1 | 127.0.0.1:8000 | FastAPI - Swagger UI | AI TASK 1 | TASK 1 AI - Google Docs

127.0.0.1:8000/docs#/default/ask_ask_post

```
date: Wed,15 Apr 2026 09:10:37 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header.

Example Value | Schema

```
"string"
```

422	Validation Error	No links

Media type

application/json

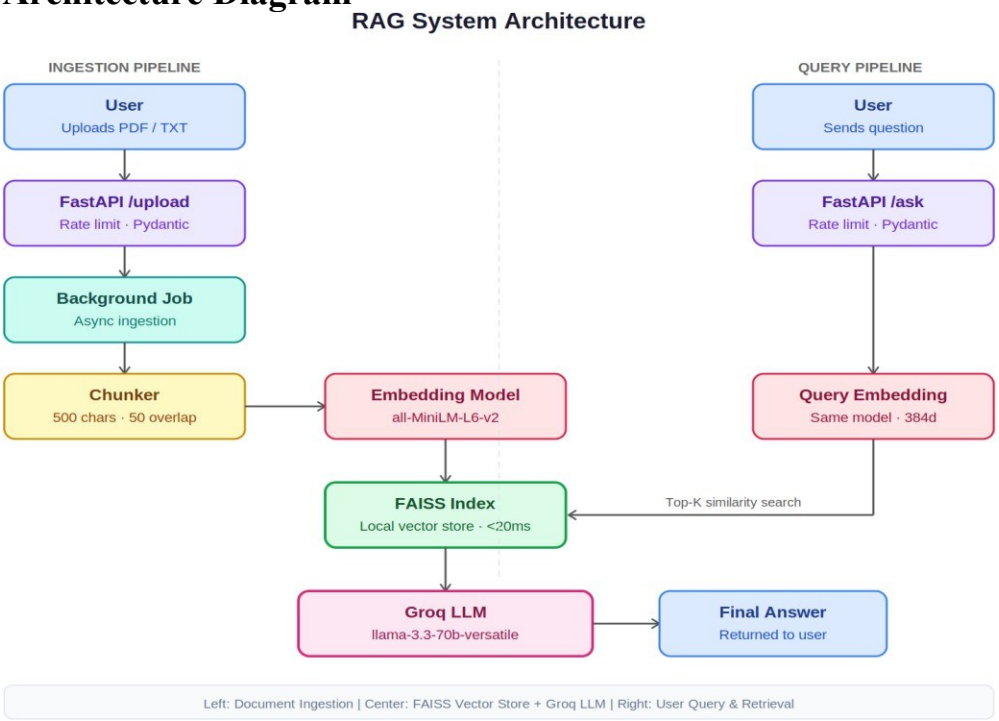
Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Schemas

Search | 25°C Clear | 19:26 15-04-2026

5. Architecture Diagram



GitHub Link:

https://github.com/AYESHA-VARSHNEY/RAG-QA-System